METHODOLOGIES TO EXPERIMENTALLY VALIDATE BBB PERMEABILITY

Presented by Brenden Park

TABLE OF CONTENTS

IN VITRO

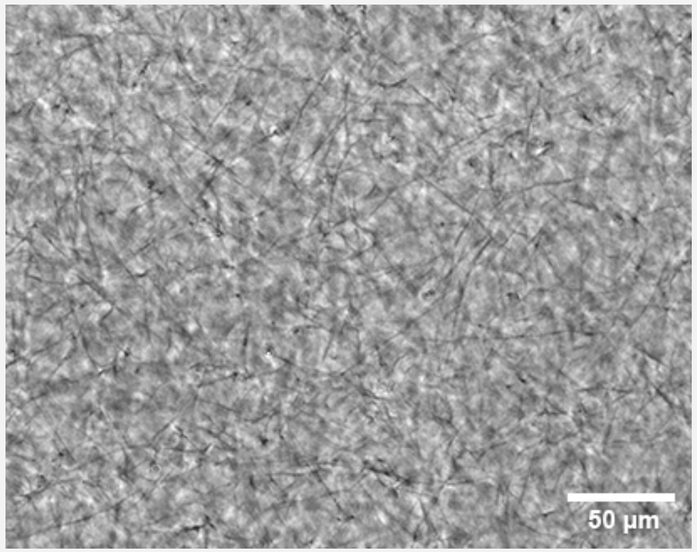
IN VIVO (ANIMAL)

IN VIVO (HUMAN)

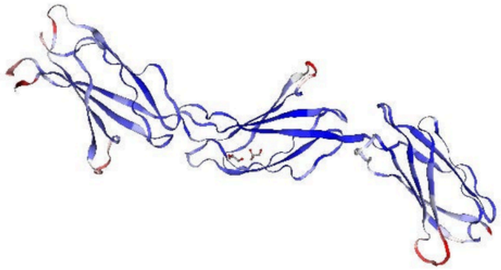
IN VITRO

(FU ET. AL, 2022 ON SPRINGER NATURE)

1. An in vitro model of the BBB is produced by layering brain endothelial cells (BECs) around plastic wells coated with rat tail collagen I and fibronectin



Rat Tail Collagen I

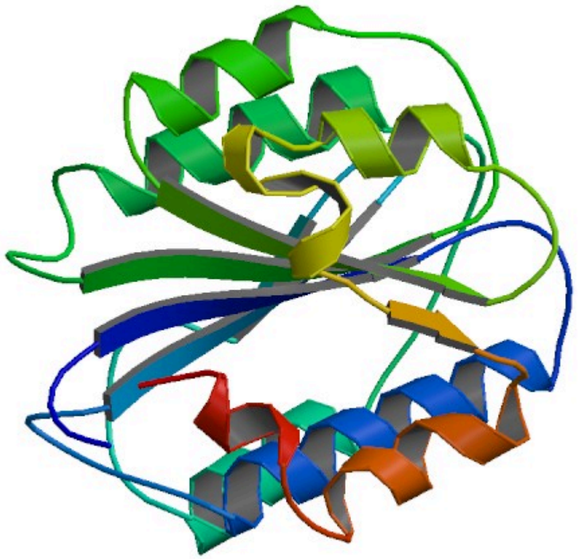


Fibronectin

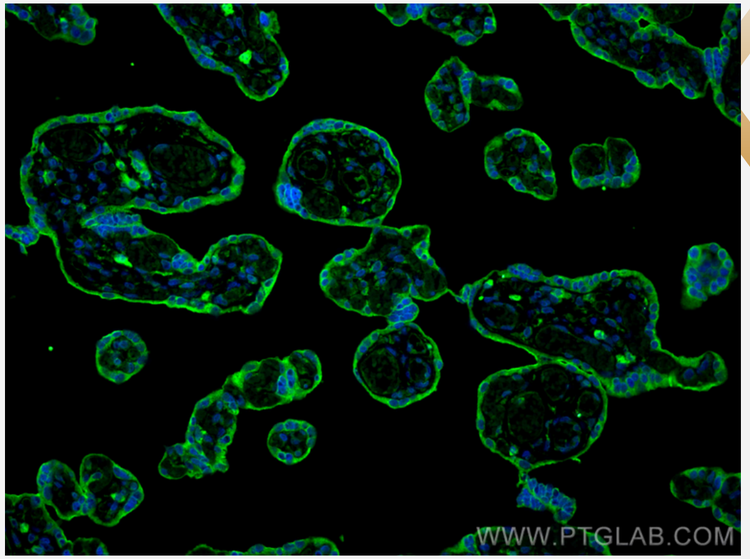
IN VITRO

(FU ET. AL, 2022 ON SPRINGER NATURE)

2. Staining was used to search for essential brain proteins such as Von Willebrand Factor (blood-clotting protein) and CD105 (receptor protein) while confirming the absence of other proteins such as SMA (smooth muscle cell actin)



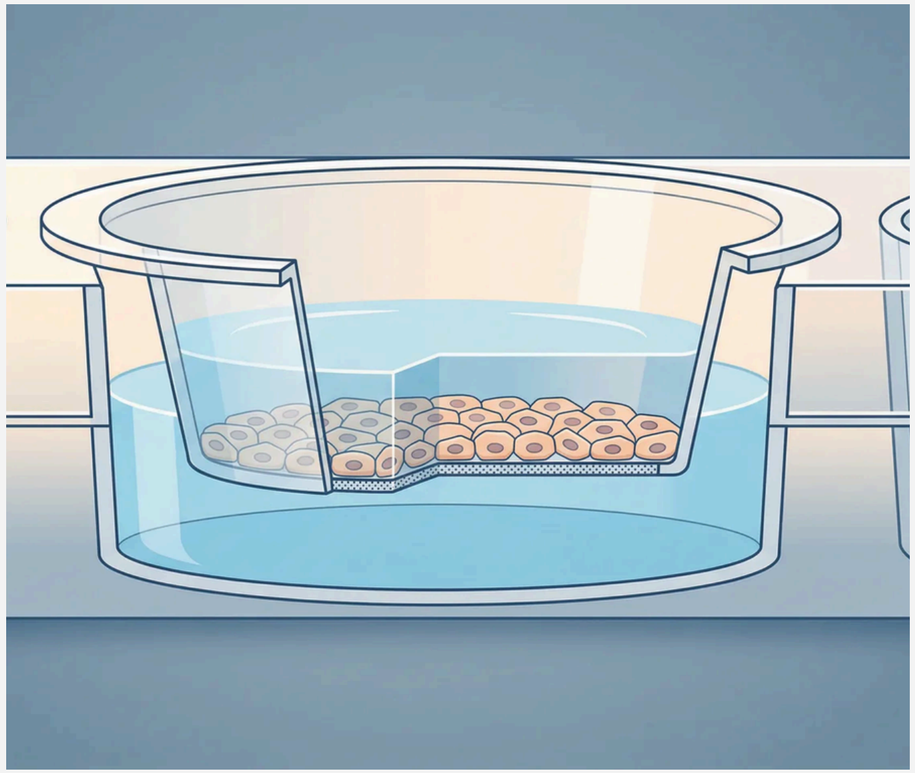
Von Willebrand Factor



CD105

IN VITRO

(FU ET. AL, 2022 ON SPRINGER NATURE)



Top and Bottom Chamber Diagram

3. Add cyclic adenosine monophosphate (cAMP) to the bottom chamber or coculture with astrocytes or pericytes to regulate and reinforce tight junctions, overall increasing the BBB's barrier function

IN VITRO

(FU ET. AL, 2022 ON SPRINGER NATURE)

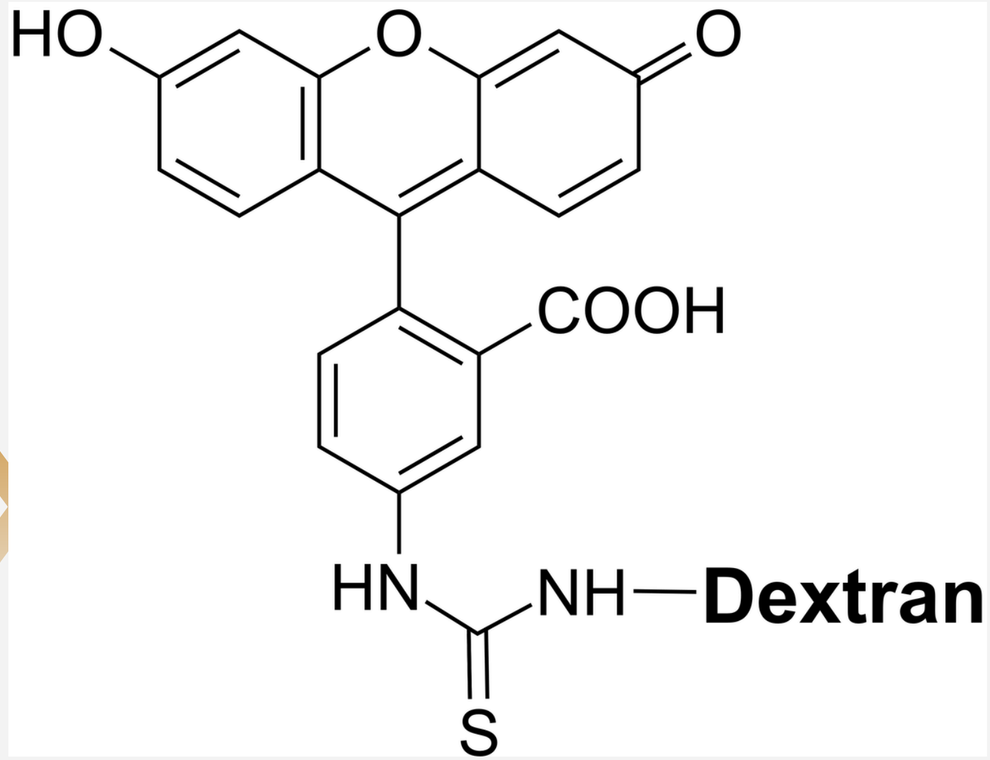
4. Add 100 microliters of PBS (Phosphate-Buffered Saline) into 8 wells of the 96-well fluorescence-based microplate to simulate the body's pH levels



Phosphate-Buffered Saline

IN VITRO

(FU ET. AL, 2022 ON SPRINGER NATURE)



FITC-Dextran

5. Add 5 microliters of FITC-dextran to the upper chamber of the well and mark it as the first time point needed to be tested. Afterwards, transfer the medium out of the upper chamber and move to its designated well

IN VITRO

(FU ET. AL, 2022 ON SPRINGER NATURE)

6. After moving the plate to the biosafety hood, collect 5 microliters of medium from the bottom chamber and transfer to designated cell, marking it as the first time point needed to be tested

7. Add 200 microliters of FITC-dextran to the insert and start the timer.

8. Collect 5 microliters of medium from both the upper and lower chamber every 10 minutes up to the 30 minute mark for permeability analysis

IN VITRO

(FU ET. AL, 2022 ON SPRINGER NATURE)

9. Use a plate reader to calculate the concentration of FITC-dextran in both the upper and lower chambers for each measured time point.



Plate Reader

IN VITRO

(FU ET. AL, 2022 ON SPRINGER NATURE)

10. Plot a ΔV_c over time graph over all time intervals, where ΔV_c is the volume cleared and is calculated as follows:

$$\Delta Vc = C_{\text{lower}} \times \frac{V_{\text{lower}}}{C_{\text{upper}}}$$

11. Obtain the value of the permeability surface area (PS) product by calculating the slope of the graph's linear regression line.

12. Use the following formula to obtain the permeability coefficient of the insert (cell + filter):

$$P_{\text{cell+filter}} = \text{PS}/s$$

13. Calculate P_{filter} by following the above steps but without the cells. Afterwards, calculate the permeability coefficient of FITC-dextran through the cells using the following formula.

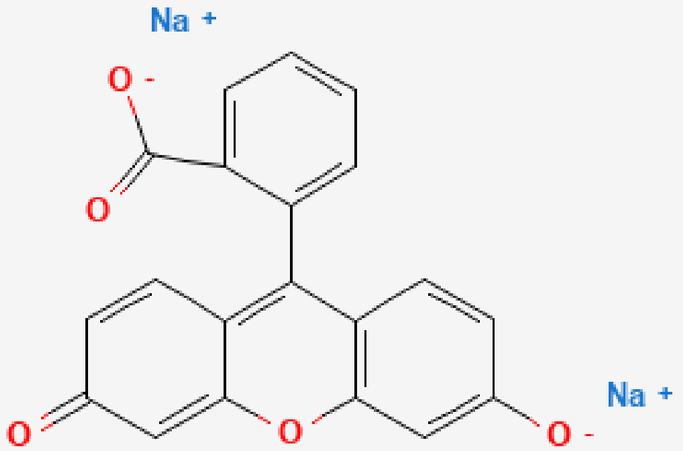
$$1/P_{\text{cell}} = 1/P_{\text{cell+filter}} - 1/P_{\text{filter}}$$

IN VIVO (ANIMAL)

(FU ET. AL, 2022 ON SPRINGER NATURE)

1. Make different concentrations of fluorescence probe solutions (NaFl or FITC-dextran) and put them into chambers

2. Adjust the optical settings of the microscope to ensure that the intensity of the fluorescence probe solution is linear with concentration



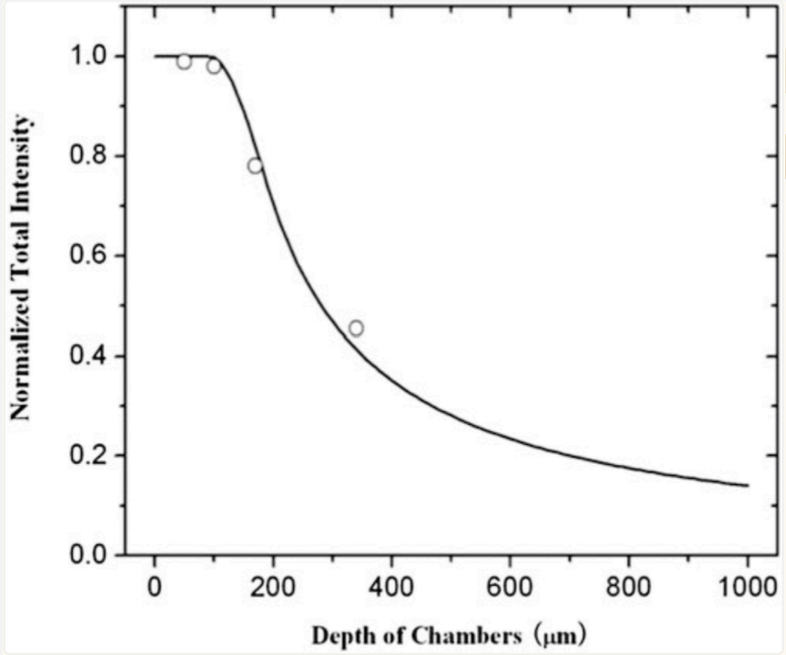
Sodium Fluorescein (NaFl)

IN VIVO (ANIMAL)

(FU ET. AL, 2022 ON SPRINGER NATURE)

3. Make four different chambers of sodium fluorescein solutions with different gap depths (50, 100, 170, 340 micrometers)

4. Use a light collection index function (mathematical model) to measure the microscope's observed light intensity for different depths



Light Collection Index Function

IN VIVO (ANIMAL)

(FU ET. AL, 2022 ON SPRINGER NATURE)

5. Put the rat under anesthesia

6. After anesthesia, shave the hair, cut the skin, and cut the connective tissue from a small region containing the rat's right or left carotid artery.

7. Cannulate the rat's right or left carotid artery to then inject fluorescent solution. This is done to be able to observe the microvessels using the microscope system

IN VIVO (ANIMAL)

(FU ET. AL, 2022 ON SPRINGER NATURE)

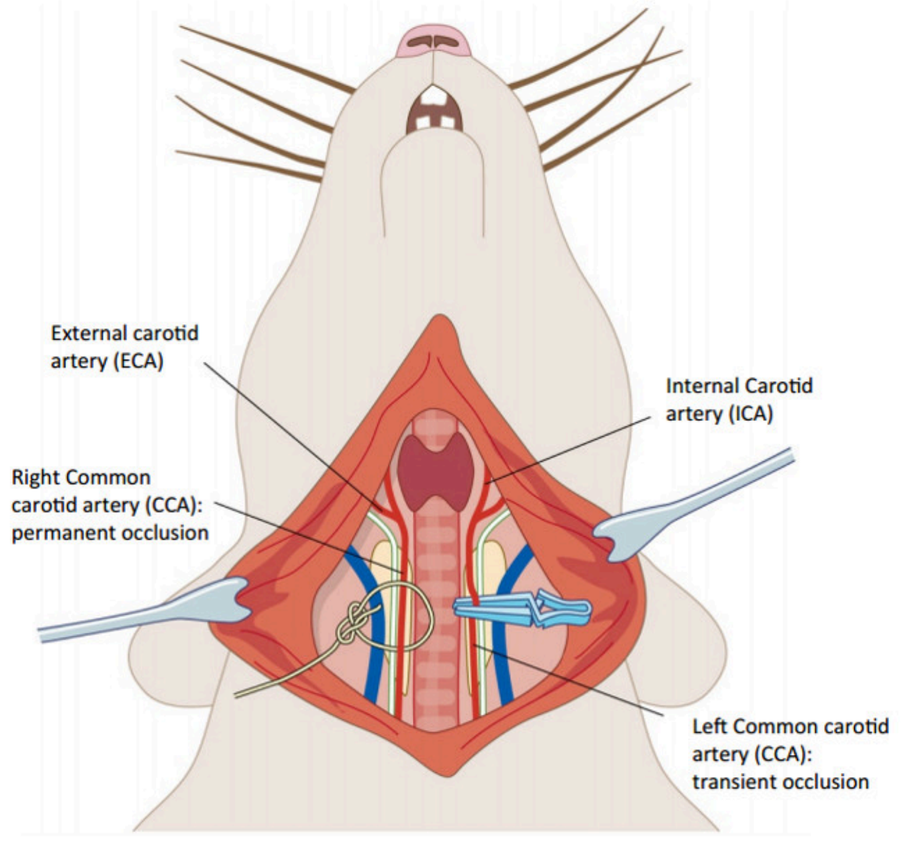


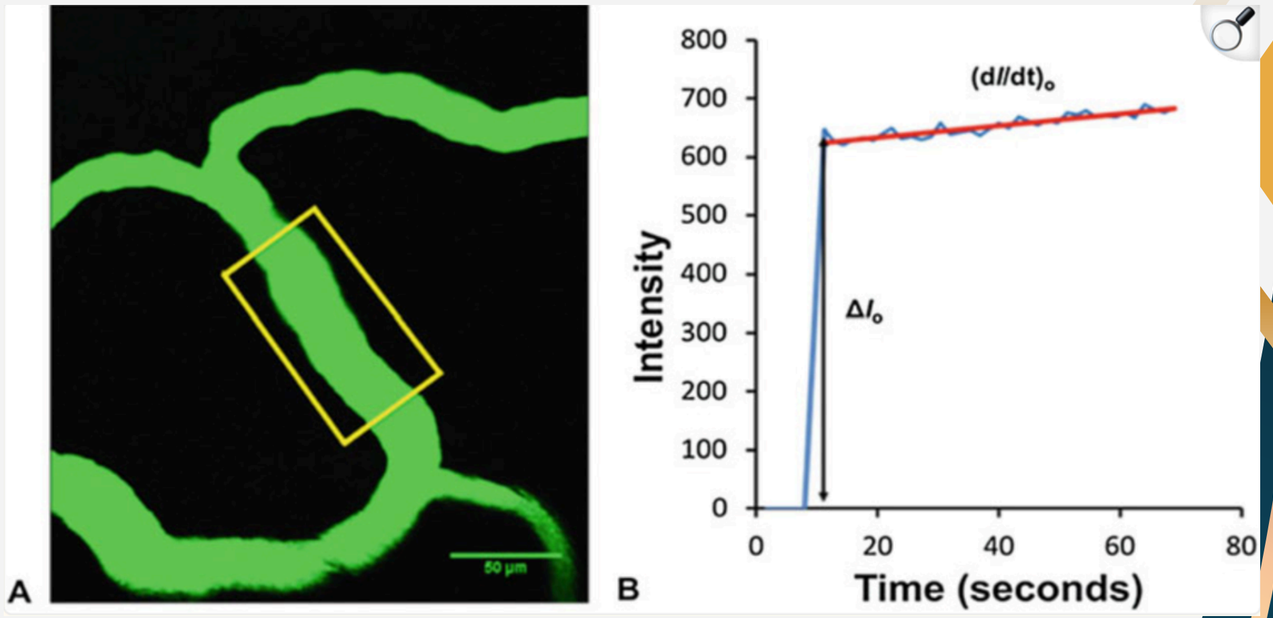
Diagram of Rat's Right and Left Cartoid Artery

IN VIVO (ANIMAL)

(FU ET. AL, 2022 ON SPRINGER NATURE)

8. Using 2D images, measure the total fluorescence intensity of a microvessel over time.

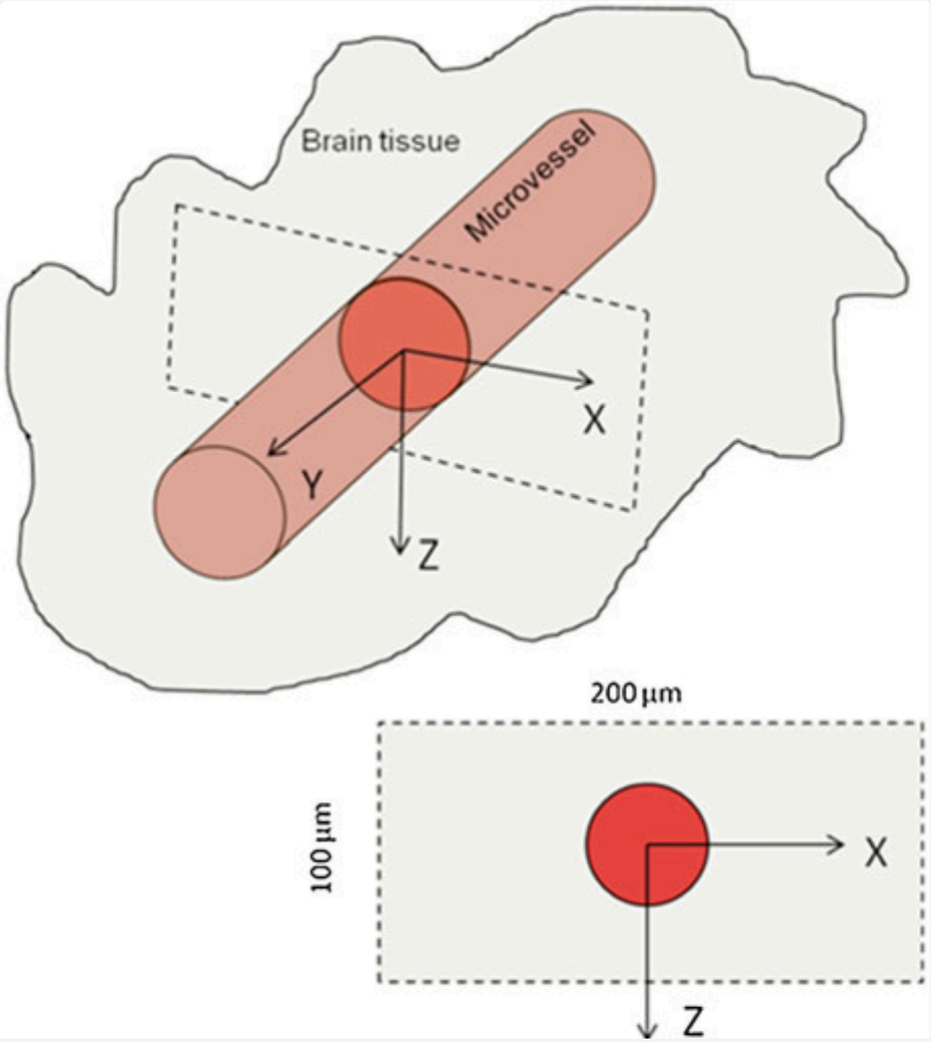
9. Calculate the solute permeability by taking the slope of the graph's regression line



A) 2D image of microvessel
B) Graph of total flourescene intensity

IN VIVO (ANIMAL)

(FU ET. AL, 2022 ON SPRINGER NATURE)

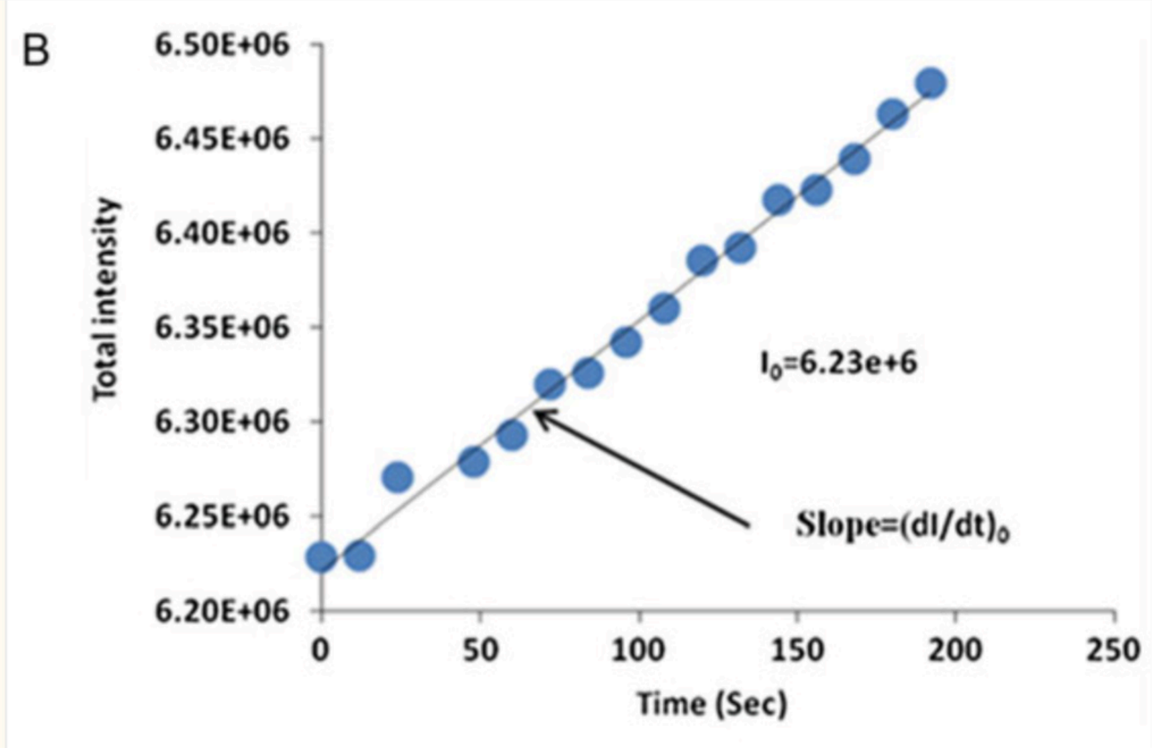


10. Use the microscope to generate a cross-sectional 3D image of the rat's microvessel containing the fluorescein solution.

IN VIVO (ANIMAL)

(FU ET. AL, 2022 ON SPRINGER NATURE)

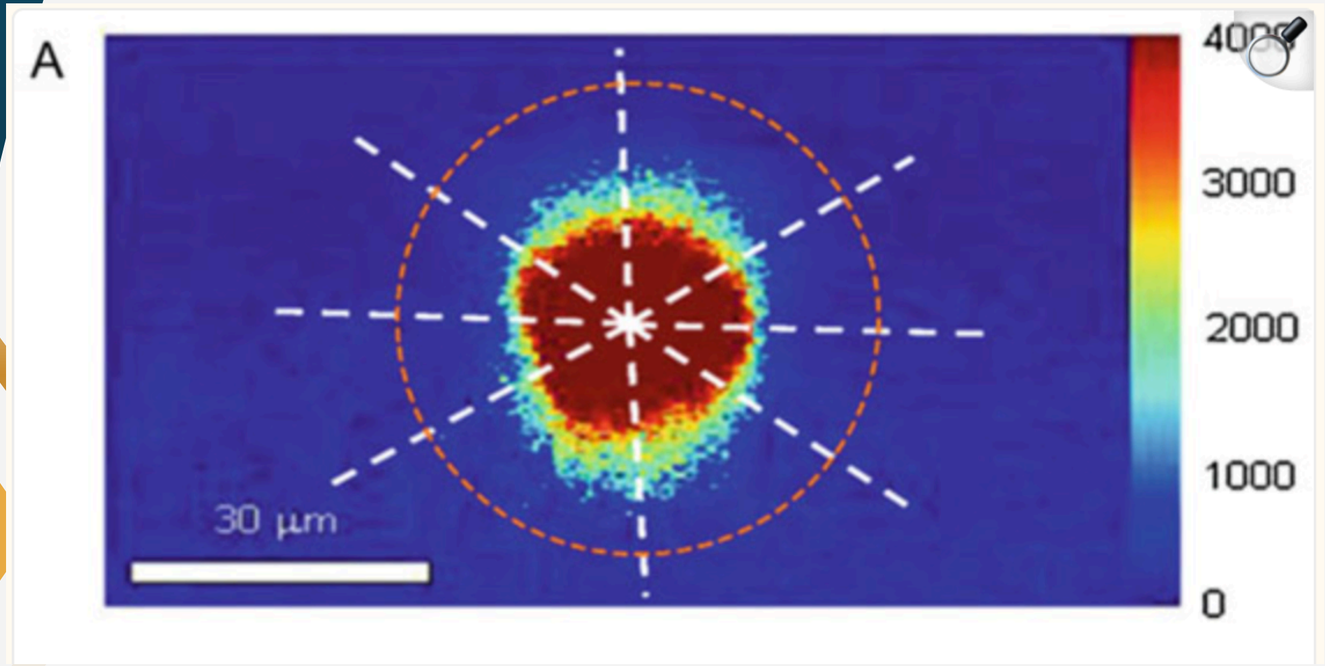
11. Measure the accumulated total intensity of the ROI (region of interest) in the 3D image as a function of time and use the slope of the regression line to calculate solute permeability.



Total Intensity Graph

IN VIVO (ANIMAL)

(FU ET. AL, 2022 ON SPRINGER NATURE)



Cross-Sectional Image

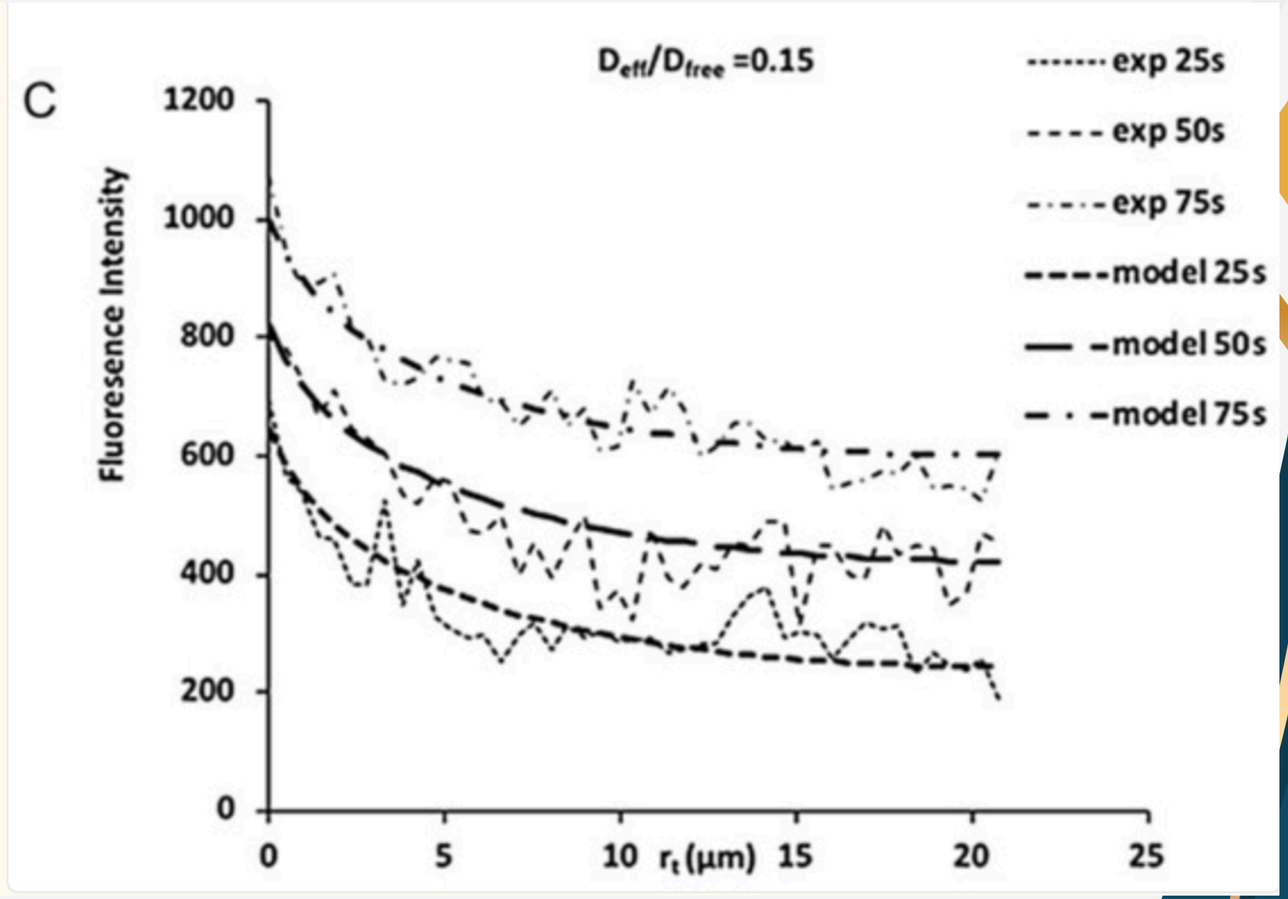
12. Draw eight straight lines in different directions from the center of the cross-sectional image

IN VIVO (ANIMAL)

(FU ET. AL, 2022 ON SPRINGER NATURE)

13. The averaged fluorescence intensity along the eight lines at different time intervals are plotted against the distance from the vessel wall.

14. The solute diffusion coefficient value that best fits the plotted curves is chosen to be the value.



Flouresence Intensity against Vessel Distance Graph

IN VIVO (ANIMAL)

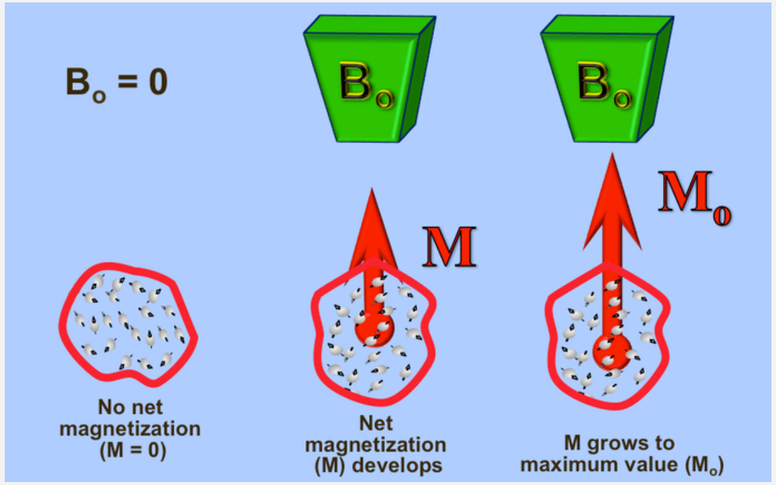
(FU ET. AL, 2022 ON SPRINGER NATURE)

15. The solute diffusion coefficient is corrected by taking into account the red blood cells, free dye, and solvent drag

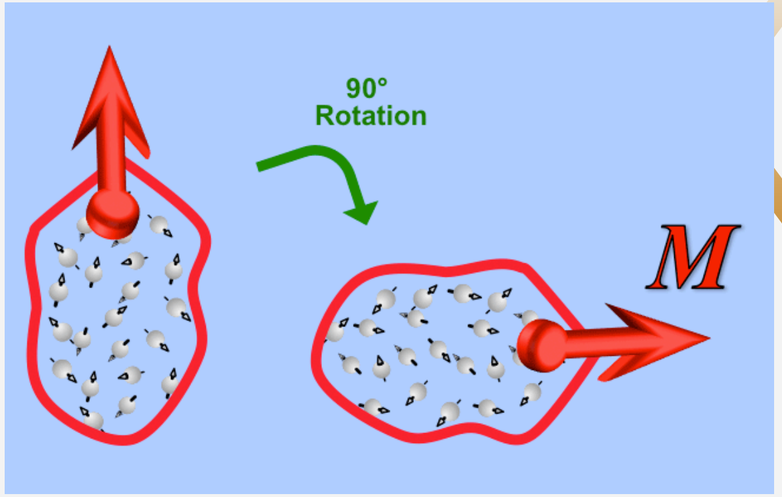
IN VIVO (HUMAN)

(CRAMER & LARSSON, 2014 ON JCBFM)

1. Magnetic resonance imaging (MRIs) were performed on each human patient to generate an array of 2D cross-sectional slices. Relaxation time and equilibrium magnetization were also measured over time intervals.



Equilibrium Magnetization



Relaxation Time

IN VIVO (HUMAN)

(FU ET. AL, 2022 ON SPRINGER NATURE)

2. A contrast agent is injected through automatic bolus injection to highlight target areas

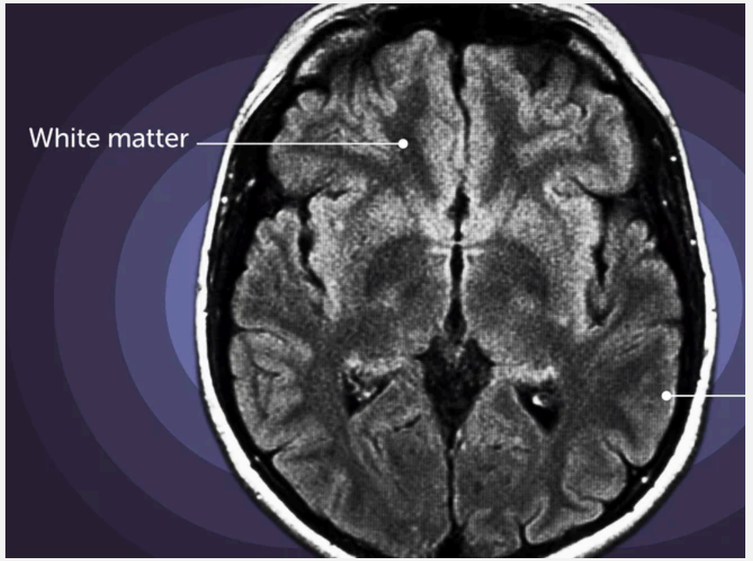


Contrast Agent

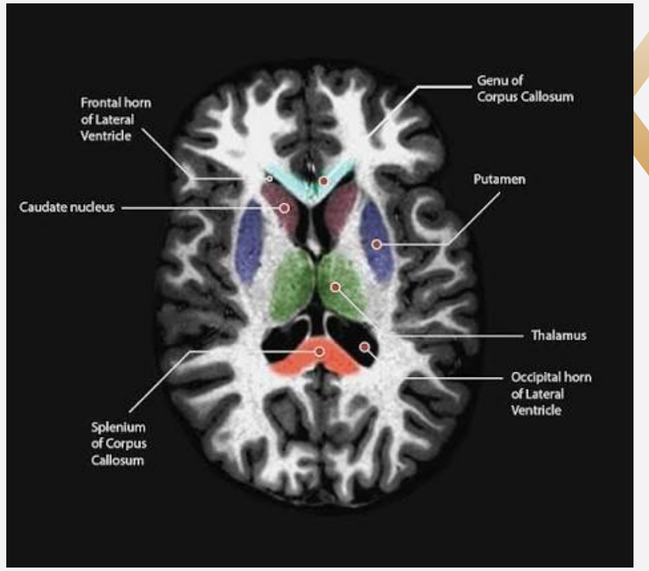
IN VIVO (HUMAN)

(FU ET. AL, 2022 ON SPRINGER NATURE)

3. The Arterial Input Function (AIF), which measures the flow of contrast dye into the artery over time, and regions of interest (ROIs) around the basal ganglia and frontal white matter were noted



Frontal White Matter



Basal Ganglia

IN VIVO (HUMAN)

(FU ET. AL, 2022 ON SPRINGER NATURE)

4. Using the DCE-MRI data obtained, MatLab-based software was used to analyze the data.

5. Simulated data of the tissue concentration curves (the total amount of a substance changes inside a specific tissue over time) were generated by using a two-compartment model having a biexponential residue impulse response function with in vivo data.

IN VIVO (HUMAN)

(FU ET. AL, 2022 ON SPRINGER NATURE)

6. The Patlak model was used by inputting generated tissue concentration curves and DCE-MRI data to obtain the blood-to-brian transfer constant value for cases with little back-diffusion from tissue to blood

$$\frac{C_t(t)}{C_a(t)} = K_i \frac{\int_0^t C_a(\tau)d\tau}{C_a(t)} + V_b$$

Patlak Model Equation

THANK YOU!

QUESTIONS?